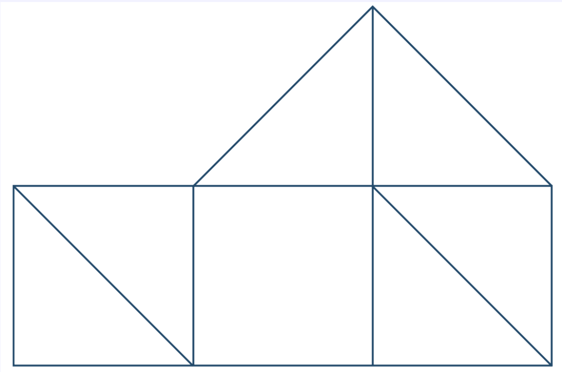
Exercise Sheet #2: DC 1: Circuit Calculations

Discussion: Oct 22, 2025

Exercise 1: Topology

Find the number of nodes, branches, and loops.

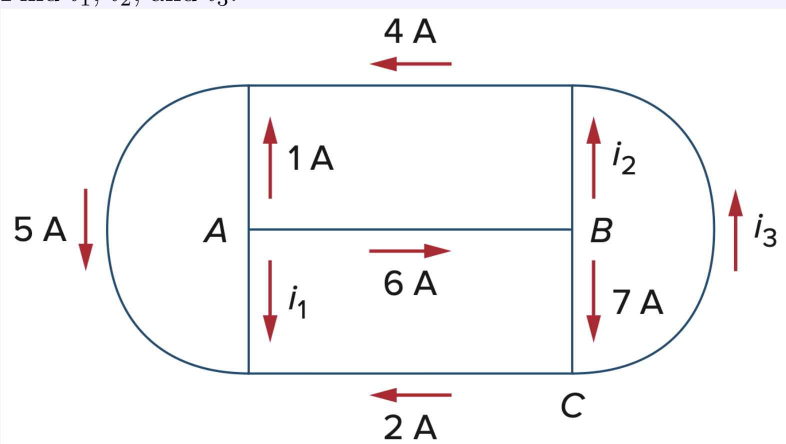


Solution

There are $n = 9$ nodes, corresponding to the junctions.
There are $b = 15$ branches, corresponding to the individual segments connecting two nodes.
There are overall $l = 15$ geometric loops, i.e., closed paths. But only $m = b - n + 1 = 7$ of these closed paths are independent loops. One could choose the 7 meshes (smallest possible loops, not enclosing any other loop) for an electrical network analysis, but other combinations are also possible.

Exercise 2: KCL

Find i_1, i_2, and i_3.



Solution

node A:

$$-1\,\mathrm{A} - 6\,\mathrm{A} - i_1 = 0$$
$$\Rightarrow i_1 = -1\,\mathrm{A} - 6\,\mathrm{A} = -7\,\mathrm{A}$$

node B:

$$6\,\mathrm{A} - 7\,\mathrm{A} - i_2 = 0$$
$$\Rightarrow i_2 = 6\,\mathrm{A} - 7\,\mathrm{A} = -1\,\mathrm{A}$$

node C:

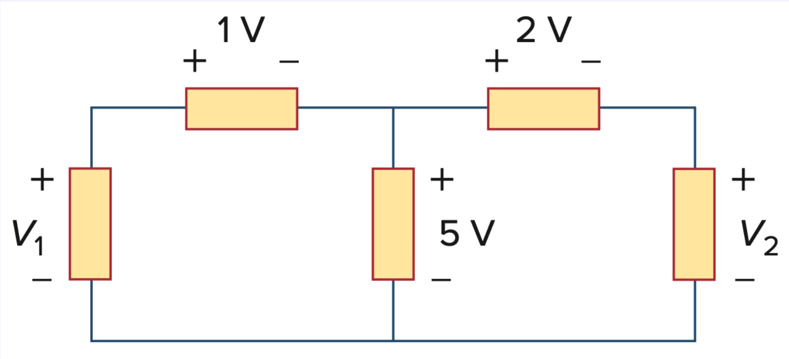
$$7\,\mathrm{A} - 2\,\mathrm{A} - i_3 = 0$$
$$\Rightarrow i_3 = 7\,\mathrm{A} - 2\,\mathrm{A} = 5\,\mathrm{A}$$

... other nodes:

$$i_2 + i_3 - 4\,\mathrm{A} = 0$$
$$\Rightarrow i_2 = 4\,\mathrm{A} - i_3 = -1\,\mathrm{A}$$

Exercise 3: KVL

Calculate V_1 and V_2.



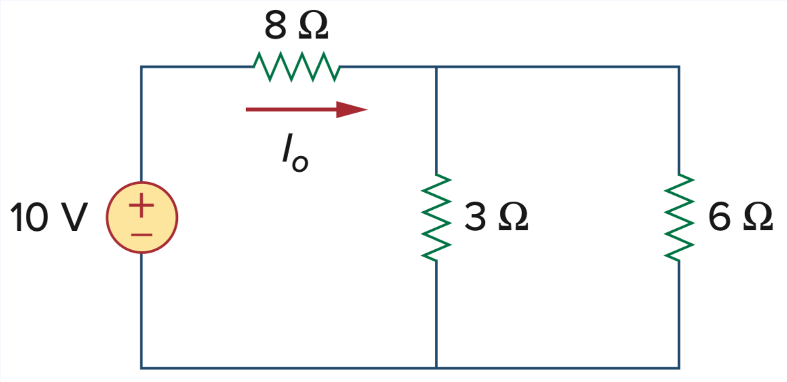
Solution

$$1\,\mathrm{V} + 5\,\mathrm{V} - V_1 = 0$$
$$\Rightarrow V_1 = 1\,\mathrm{V} + 5\,\mathrm{V} = 6\,\mathrm{V}$$

$$2\,\mathrm{A} + V_2 - 5\,\mathrm{V} = 0$$
$$\Rightarrow V_2 = 5\,\mathrm{V} - 2\,\mathrm{V} = 3\,\mathrm{V}$$

Exercise 4: Series and Parallel Resistors

Calculate I_0.



Solution

First, we calculate the equivalent resistance of the parallel resistors with $3\,\Omega$ and $6\,\Omega$, then we add the series resistor with $8\,\Omega$:

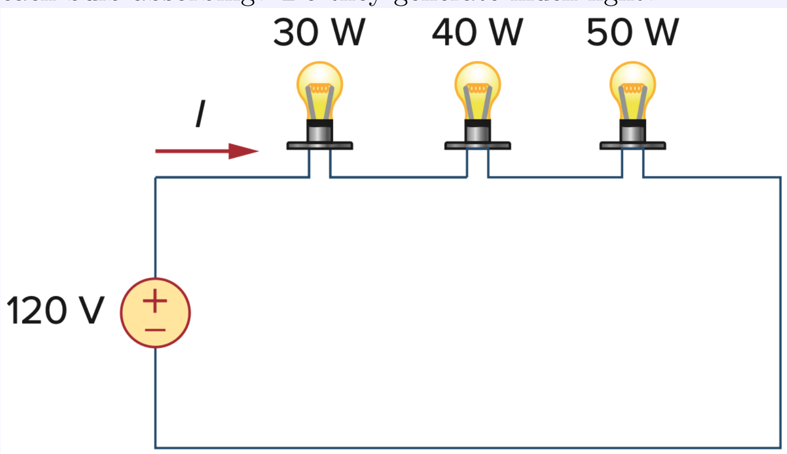
$$R = \frac{3\,\Omega \cdot 6\,\Omega}{3\,\Omega + 6\,\Omega} + 8\,\Omega = 10\,\Omega$$

Finally, we can calculate the current I_0 using Ohm's law:

$$I_o = \frac{10\,\mathrm{V}}{10\,\Omega} = 1\,\mathrm{A}$$

Exercise 5: Series vs. Parallel

Three light bulbs are connected in series to a $120\,\mathrm{V}$ source as shown below. Find the current I through each of the bulbs. Each bulb is rated at $120\,\mathrm{V}$. How much power is each bulb absorbing? Do they generate much light?



Solution

First, we calculate the resistances of each bulb using their rated power and voltage:

$$R = \frac{V^2}{P}$$

$$R_{30\,\mathrm{W}} = \frac{(120\,\mathrm{V})^2}{30\,\mathrm{W}} = 480\,\Omega$$

$$R_{40\,\mathrm{W}} = \frac{(120\,\mathrm{V})^2}{40\,\mathrm{W}} = 360\,\Omega$$

$$R_{50\,\mathrm{W}} = \frac{(120\,\mathrm{V})^2}{50\,\mathrm{W}} = 288\,\Omega$$

Then, we calculate the equivalent resistance of the three light bulbs in series:

$$R = 480\,\Omega + 360\,\Omega + 288\,\Omega = 1128\,\Omega$$

Then, we can calculate the current I using Ohm's law:

$$I = \frac{120\,\mathrm{V}}{1128\,\Omega} = 0.1063\,\mathrm{A}$$

Finally, we can calculate the power absorbed by each bulb:

$$P = I^2 \cdot R$$

$$P_{30\,\mathrm{W}} = (0.1063\,\mathrm{A})^2 \cdot 480\,\Omega = 5.432\,\mathrm{W}$$

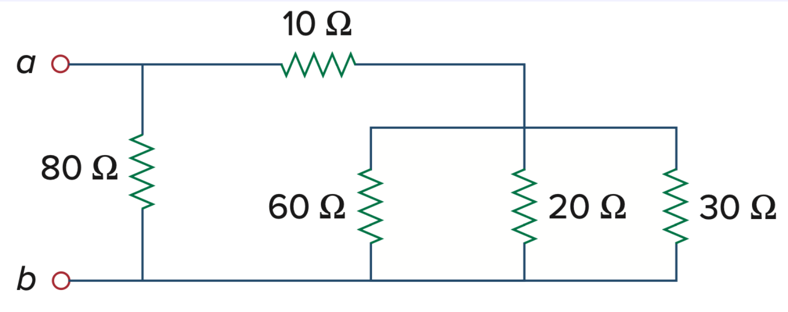
$$P_{40\,\mathrm{W}} = (0.1063\,\mathrm{A})^2 \cdot 360\,\Omega = 4.074\,\mathrm{W}$$

$$P_{50\,\mathrm{W}} = (0.1063\,\mathrm{A})^2 \cdot 288\,\Omega = 3.259\,\mathrm{W}$$

Clearly these values are well below the rated powers of each light bulb, so we would not expect very much light from any of them. To work properly, they need to be connected in parallel.

Exercise 6: Equivalent Resistance

Calculate the equivalent resistance R_{ab} across the terminals.



Solution

First we calculate the equivalent resistance of the parallel resistors with $60\,\Omega$, $20\,\Omega$, and $30\,\Omega$:

$$R_p = \frac{1}{\frac{1}{60\,\Omega} + \frac{1}{20\,\Omega} + \frac{1}{30\,\Omega}} = 10\,\Omega$$

Then we add the series resistor with $10\,\Omega$:

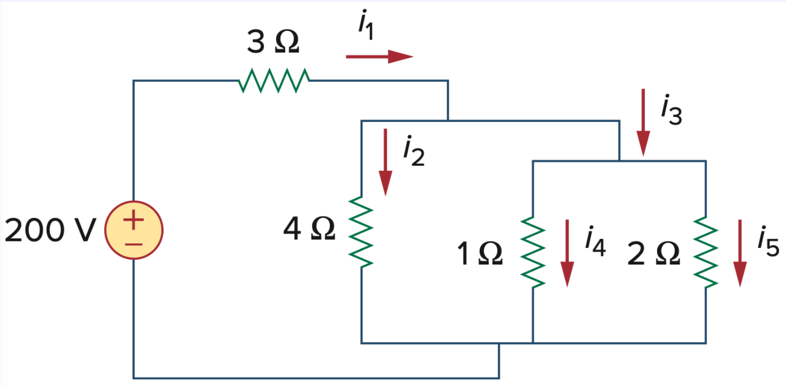
$$R_{ab} = 10\,\Omega + 10\,\Omega = 20\,\Omega$$

And finally we combine with the parallal resistor of $80\,\Omega$:

$$R_{ab} = \frac{1}{\frac{1}{20\,\Omega} + \frac{1}{80\,\Omega}} = 16\,\Omega$$

Exercise 7: Current Divider Rule

Calculate i_1 to i_5.



Solution

First, we calculate the equivalent resistance of the parallel resistors with $1\,\Omega$ and $2\,\Omega$ and call it R_3:

$$R_3 = \frac{1\,\Omega \cdot 2\,\Omega}{1\,\Omega + 2\,\Omega} = \frac{2}{3}\,\Omega$$

Then we calculate the total resistance R from the $4\,\Omega$ resistor parallel to R_3 and the $3\,\Omega$ resistor in series:

$$R = 3\,\Omega + \frac{4\,\Omega \cdot \frac{2}{3}\,\Omega}{4\,\Omega + \frac{2}{3}\,\Omega} = \frac{25}{7}\,\Omega$$

Using Ohm's law, we can calculate the total current i_1:

$$i_1 = \frac{200\,\mathrm{V}}{\frac{25}{7}\,\Omega} = 56\,\mathrm{A}$$

Then, we can calculate i_2 using current division:

$$i_2 = i_1 \cdot \frac{R_3}{R_2 + R_3} = 56\,\mathrm{A} \cdot \frac{\frac{2}{3}\,\Omega}{4\,\Omega + \frac{2}{3}\,\Omega} = 8\,\mathrm{A}$$

KCL gives us i_3:

$$i_3 = i_1 - i_2 = 56\,\mathrm{A} - 8\,\mathrm{A} = 48\,\mathrm{A}$$

Again, we can calculate i_4 using current division:

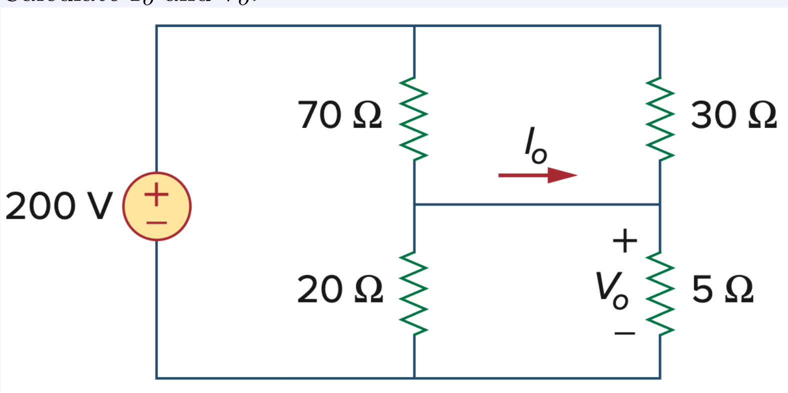
$$i_4 = i_3 \cdot \frac{R_5}{R_4 + R_5} = 48\,\mathrm{A} \cdot \frac{2\,\Omega}{1\,\Omega + 2\,\Omega} = 32\,\mathrm{A}$$

Finally, KCL gives us i_5:

$$i_5 = i_3 - i_4 = 48\,\mathrm{A} - 32\,\mathrm{A} = 16\,\mathrm{A}$$

Exercise 8: Node Connection

Calculate I_o and V_o.



Solution

A node is a point where two or more circuit elements connect (electrically the same point). A branch is a path that connects two nodes and contains a circuit element (resistor, source, etc.).
Here, there are three main nodes:

- Top node: connected to the +200 V source, top of the 70 Ω and 30 Ω resistors.
- Bottom node: 0 V reference.
- Middle node: connection between two branches. On the left, the branch formed by the 70 Ω and 20 Ω. And on the right, the branch formed by the 30 Ω and 5 Ω.

The middle horizontal wire (with I_o) is part of the middle node, but because there's a potential difference between the left and right parts of that node before it's connected, current flows through that connection — hence we analyze I_o as if it's a branch current.
As the output voltage V_o is the voltage across the 5 Ω resistor, the output voltage equals the middle node voltage.

Currents from the top node to the middle node:

$$I_{70} = \frac{200\,\mathrm{V} - V_o}{70\,\Omega}, \qquad I_{30} = \frac{200\,\mathrm{V} - V_o}{30\,\Omega}$$

Currents from the middle node to the bottom node:

$$I_{20} = \frac{V_o}{20\,\Omega}, \qquad I_5 = \frac{V_o}{5\,\Omega}$$

Applying KCL at the middle node:

$$\frac{200\,\mathrm{V} - V_o}{70\,\Omega} + \frac{200\,\mathrm{V} - V_o}{30\,\Omega} = \frac{V_o}{20\,\Omega} + \frac{V_o}{5\,\Omega}$$

Simplify:

$$\frac{200\,\mathrm{V} - V_o}{21\,\Omega} = \frac{V_o}{4\,\Omega}$$

$$\Rightarrow 800\,\mathrm{V} - 4 \cdot V_o = 21 \cdot V_o$$

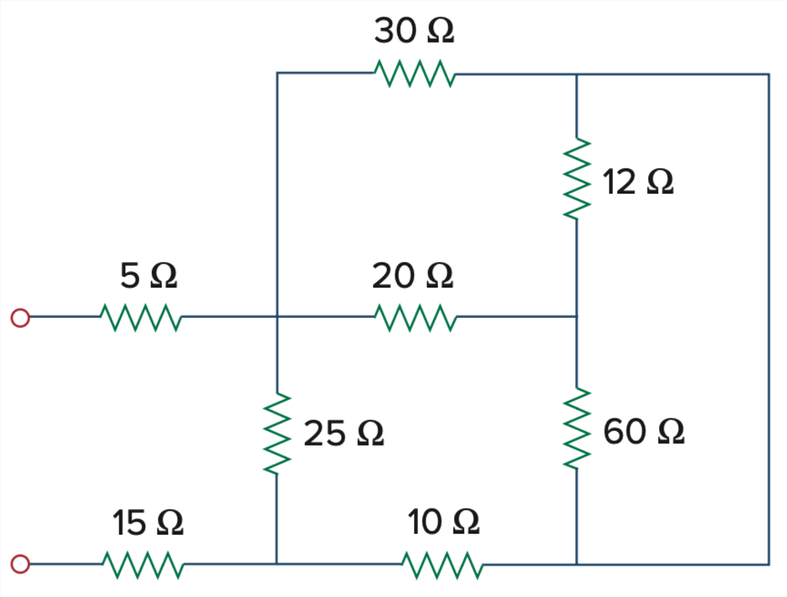
$$\Rightarrow \boxed{V_o = 32\ \mathrm{V}}$$

Now we use KCL to compute the current I_o (defined to the right):

$$I_o = I_{70} - I_{20} = \frac{200\,\mathrm{V} - 32\,\mathrm{V}}{70\,\Omega} - \frac{32\,\mathrm{V}}{20\,\Omega} = \boxed{0.8\,\mathrm{A}}$$

Exercise 9: Δ-Y Transformation

Calculate the equivalent resistance across the terminals.


30 Ω
12 Ω
5 Ω
20 Ω
25 Ω
60 Ω
15 Ω
10 Ω

Solution

1. Combine the right vertical resistors:

$$12\,\Omega \parallel 60\,\Omega = \frac{12\,\Omega \cdot 60\,\Omega}{12\,\Omega + 60\,\Omega} = 10\,\Omega$$

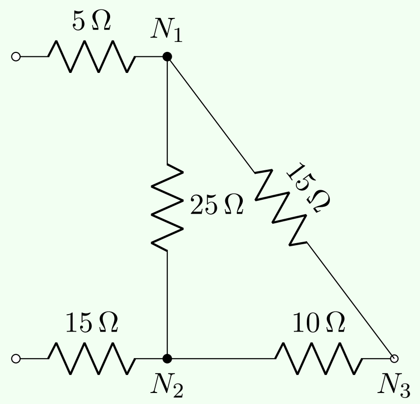
2. Series combination with the $20\,\Omega$ resistor:

$$20\,\Omega + 10\,\Omega = 30\,\Omega$$

3. Parallel combination with the other $30\,\Omega$:

$$30\,\Omega \parallel 30\,\Omega = \frac{30\,\Omega \cdot 30\,\Omega}{30\,\Omega + 30\,\Omega} = 15\,\Omega$$

After these three steps, the circuit reduces to a triangle (Δ):

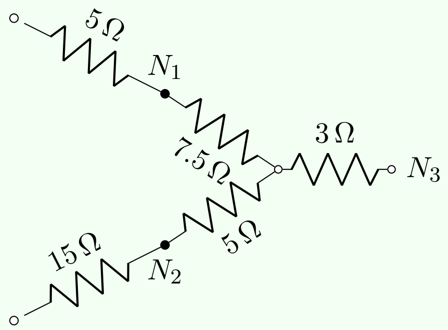


4. Convert the Δ-network to equivalent Y-resistances:

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} = \frac{15\,\Omega \cdot 25\,\Omega}{50\,\Omega} = 7.5\,\Omega$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c} = \frac{10\,\Omega \cdot 25\,\Omega}{50\,\Omega} = 5\,\Omega$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{10\,\Omega \cdot 15\,\Omega}{50\,\Omega} = 3\,\Omega$$



The branch with the $3\,\Omega$ resistor connects to a dead-end node N_3, so it carries no current and can be ignored.

$$\Rightarrow R_{eq} = 5\,\Omega + 7.5\,\Omega + 5\,\Omega + 15\,\Omega = 32.5\,\Omega$$